

HOMEWORK #2

(Due: Wednesday, March 1, 2023, 1:00 pm PT via Brightspace)

1. **Reconstruction.** Consider a discrete time signal $x(n)$ with a sampling rate of $f_s = 125$ MHz that is converted to a continuous time representation using zero-order-hold pulses.
 - (a) Plot the amplitude envelope of the continuous-time signal resulting from hold pulse widths of $T_{p1} = 2\text{ns}$, $T_{p2} = 4\text{ns}$ and $T_{p3} = 8\text{ns}$, all on the same graph. Leave the x-axis in absolute frequency rather than normalizing it to f_s .
 - (b) Suppose that the discrete time signal $x(n)$ represents a 30 MHz sine wave with amplitude A . For the 8ns hold case, calculate the amplitudes (normalized to A) of all tones present in the output spectrum up to 250 MHz.
2. **Segmented DAC Design.** Read the following paper: C.-H. Lin and K. Bult, “A 10-b, 500-MSample/s CMOS DAC in 0.6 μm^2 ,” IEEE J. Solid-State Circuits, vol. 33, pp. 1948-1958, Dec. 1998. What application is this DAC intended for?
3. In the above reference, it was found that for the given objectives, a thermometer/binary partition of 8/2 is most suitable. Your task in this problem is to re-optimize this DAC for a 65-nm technology with a different set of process parameters, without altering its basic structure. The design variables to be optimized are:

Parameter	Description
B_t	Number of bits in thermometer section
A_{unit}	Area of unit current source

Your target specs and process parameters are:

Parameter	Description	Value
B	Resolution	12 bits
INL_{spec}	Worst case integral nonlinearity	< 0.5 LSB
DNL_{spec}	Worst case differential nonlinearity	< 0.5 LSB
Y	Target yield	95%
A_{decode}	Approximate decoder and routing area	$2^{B_t} \cdot 2000 \mu\text{m}^2$
A_{total}	Total DAC area (to be minimized)	$2^B \cdot A_{\text{unit}} + A_{\text{decode}}$
k_u	Unit element matching parameter	6% μm
σ_u	Standard deviation of unit element mismatch	$k_u / \sqrt{A_{\text{unit}}}$

- (a) Following the procedure outlined in the reference, answer the following **using only hand calculations** (After the equations are written by hand, it is ok to use Matlab/Excel to perform a sweep across **integer** values of B_t):
 - i. What values of A_{unit} and B_t actually minimize the total area defined above in the table? What is the minimum total area (in mm^2)?
 - ii. What values of A_{unit} and B_t give the “optimal point” described by the paper? What is the total area (in mm^2) in this case

- (b) Create a plot similar to Fig. 9 of the reference paper to illustrate the “minimum area” design. In this diagram, plot the total DAC area in mm^2 (log scale) versus the design parameter B_t (linear scale in bits).
4. Use Matlab to simulate your “minimum area” design from problem 3 part (a) i, and make appropriate modifications to achieve the targeted yield.
- (a) Download the starter file **hw2_dac.m** from the “homework” section of the course website. This file contains code for a 100-run Monte Carlo simulation that generates an INL envelope and RMS plot (see Fig. 7 and Fig. 8 of the paper). Below the line “Your code goes here,” add additional code to generate the respective plots also for DNL. Turn in a printout of the code you added to the file (no need to submit a printout of the given code).
- (b) Generate and turn in envelope and RMS plots for both DNL and INL using the parameters you calculated in your hand analysis. Note that you should use the A_{unit} and B_t values calculated above in problem 3.
- (c) Does your initial design meet the yield spec? Explain why it does or why it doesn't. Note that meeting the yield spec requires that “on average” no more than 5 out of the 100 runs lie outside the specified DNL or INL.
- (d) If your hand design does not meet the yield spec, make appropriate adjustments (OK to iterate in Matlab) to A_{unit} without changing anything else. What is the value of A_{unit} that achieves the desired yield? What is the final total area of your design?